9.Implement the image scaling techniques to resize the images to both bigger and small sizes

Aim

To write a Python program to resize an image into bigger and smaller sizes using OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the input image.
3. Check whether the image is loaded successfully.
4. Increase the image dimensions using cv2.resize().
5. Decrease the image dimensions using cv2.resize().
6. Display the original, enlarged, and reduced images.

Code

```
import cv2

img = cv2.imread("sample.jpg")

if img is None:
    print("Image not found")
else:
    bigger = cv2.resize(
        img, None, fx=2, fy=2,
        interpolation=cv2.INTER_LINEAR
    )

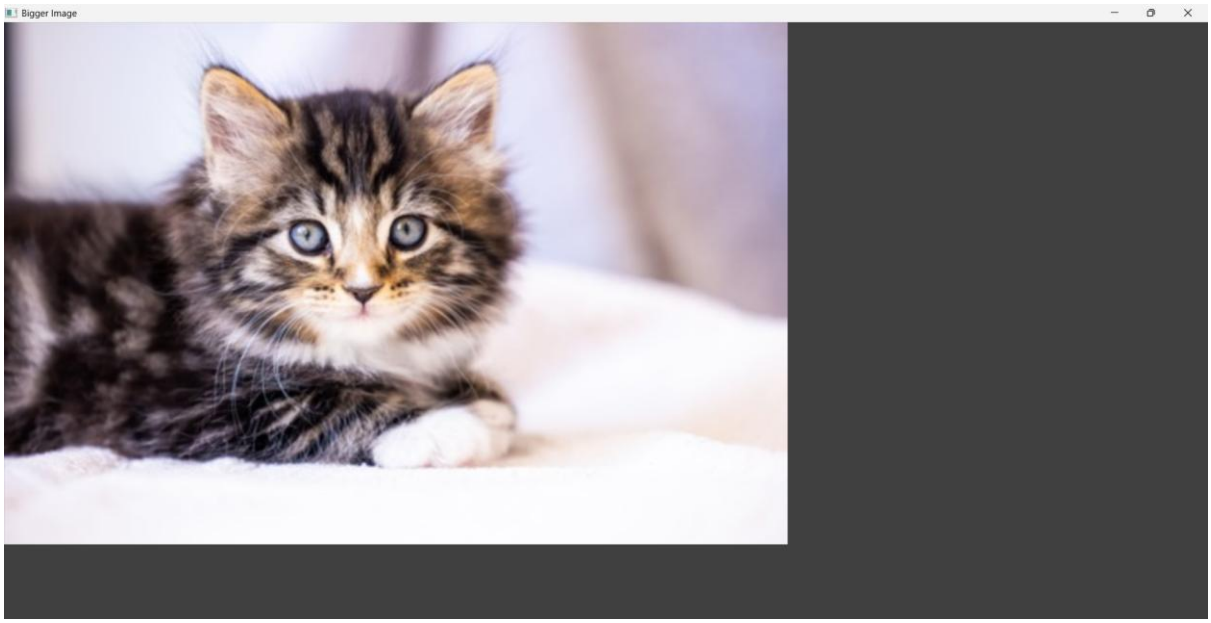
    smaller = cv2.resize(
        img, None, fx=0.5, fy=0.5,
        interpolation=cv2.INTER_AREA
    )

    cv2.imshow("Original Image", img)
    cv2.imshow("Bigger Image", bigger)
```

```
    cv2.imshow("Smaller Image", smaller)


    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

SAMPLE INPUT:



SAMPLE OUTPUT:



Result

Thus, image dilation was successfully performed using the cv2.dilate() function. The white regions in the binary image became larger.